4.4: Norms and Analysis of Errors

1) Show that the norms $\|x\|_\infty, \|x\|_2$, and $\|x\|_1$ satisfy the postulates (1), (2), and (3) for norms.

2) Show that $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ for all $x \in \mathbb{R}^n$, and that equalities can occur, even for nonzero vectors.

3) (Continuation) Show that $\|x\|_1 \leq n\|x\|_\infty$ and $\|x\|_2 \leq \sqrt{n}\|x\|_\infty$ for $x \in \mathbb{R}^n$.

8) Define

$$\|A\| = \sum_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}|$$

Show that this is a matrix norm (that is, a norm on the linear space of all $n \times n$ matrices).

Show that it is not subordinate to any vector norm. Does it conform to Equation (9) and Inequality (10)?

$$\|I\| = 1 \qquad \text{(Equation 9)}$$

$$\|AB\| \leq \|A\|\|B\| \qquad \text{(Inequality 10)}$$

11) Show that for the vector norm $\|x\|_1$ defined in Equation (5), the subordinate matrix norm is

$$\|A\|_1 = max_{1\leq j\leq n} \sum_{i=1}^{n} |a_{ij}|$$

$$\|x\|_1 = \sum_{i=1}^{n} |x_i| \qquad \text{(Equation 5)}$$

33) For any $n \times n$ matrix A, define

$$\|A\|_F = \left(\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2 \right)^{\frac{1}{2}}$$

(This is called the **Frobenius norm**.) Is this a subordinate matrix norm? Answer the same question for $\|A\| = max_{1\leq i,j\leq n}|a_{ij}|$. Prove that these equations define norms on the vector space of all $n \times n$ matrices.

4.6: Solution of Equations by iterative Methods

1b) Program the Gauss-Seidel method and test it on the following

$$\begin{cases} 3x + y + z &= 5 \\ 3x + y - 5z &= -1 \\ x + 3y - z &= 3 \end{cases}$$

Analyze what happens when these systems are solved by simple Gaussian elimination without pivoting.

6.1: Norms and Analysis of Errors

1d) Find the polynomial of least degree that interpolate these sets of data:

x	3	7	1	2
y	12	146	2	1

8) Discuss the problem of determining a polynomial of degree at most 2 for which $p(0), p(1)$, and $p'(\xi)$ are prescribed, ξ being any preassigned point.